

YEYOUNG LEE

GRADUATE RESEARCH APPLICANT

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Yong In University | Artificial Intelligence | GPA 3.82/4.5

B.S. expected Feb 2027 | Graduate applicant

Interested in reliable detection and explanation of anomalies in industrial data. My research interests focus on smart manufacturing and semiconductor process data, with particular attention to class imbalance, robust validation, explainable AI (XAI), and deployable multimodal anomaly detection.

RESEARCH INTERESTS Industrial AI · Smart Manufacturing · Anomaly Detection · Explainable AI · Multimodal Learning

EDUCATION

Yong In University, Department of Artificial Intelligence

Mar 2025 – Feb 2027 (expected)

B.S. candidate · GPA 3.82/4.5 · Transfer admission

- Relevant coursework: Artificial Intelligence, Machine Learning, Big Data Analytics, Financial Time Series, Computer Architecture, Operating Systems, Computer Networks, and Text Mining

Yong-In Arts & Science University, Department of Big Data Management

Mar 2022 – Feb 2025

Prior institution before transfer · GPA 4.46/4.5

- Completed foundational training in data analytics, management information systems, and applied statistics

RESEARCH EXPERIENCE

Fair-Value Estimation and Anomalous Transaction Detection in Seoul Real Estate

2026

Oral presentation, 59th Spring Conference of the Korea Institute of Information and Communication Engineering · [GitHub Repository](#)

- Cleaned and modeled 189,864 Seoul real-estate transactions using XGBoost regression
- Achieved mean R^2 of 0.9172 and identified 922 anomalous transaction candidates from prediction residuals
- Contributed to data analysis and result interpretation; delivered a seven-minute oral presentation and Q&A

Undergraduate Researcher / Research Intern

Jul 2023 – Feb 2024

- Conducted literature review, data collection and preprocessing, exploratory analysis, result synthesis, and research documentation

SELECTED PROJECTS

GreenFab Insight — Semiconductor Defect Prediction, Explanation, and ESG Dashboard

Aug 2026

EST AI Challengers Hackathon · Team Lead / Modeling Lead

- Designed a manufacturing-defect prediction pipeline using UCI SECOM data: 1,567 samples, 591 sensor variables, and 104 defects
- Applied LightGBM, Optuna, SHAP, balanced class weights, Stratified 5-Fold cross-validation, and decision-threshold optimization
- Evaluated with Recall, Precision, PR-AUC, Balanced Accuracy, and failure analysis rather than relying on Accuracy alone

SafeNavi — Location-Based Natural-Disaster Shelter Guidance

Mar 2026 – Jun 2026

Capstone Project · Data Collection, Preprocessing, and Database

- Collected and cleaned 87,767 nationwide shelter records and 645 disaster-action guides; built a PostgreSQL/PostGIS database
- Designed spatial data structures for nearby-shelter search and integrated them into the deployed service at [www.safenavi.site](#)

Impact Validation and Investment Recommendation for the Local Extinction Response Fund

Aug 2026 – Present

Chungbuk Regional AI Innovation Project · Team Lead

- Designed an impact-validation framework using outcome variables such as youth net migration to address the limitations of execution-rate-only evaluation
- Specified baselines, out-of-time validation, error analysis, and evidence grading based on difference-in-differences
- Planned a service flow from conversational query to diagnostic dashboard and next-year investment-plan generation

TECHNICAL SKILLS

Programming & Data Python, SQL, Pandas, NumPy, scikit-learn	Machine Learning XGBoost, LightGBM, CatBoost, Optuna, SHAP
Database & Backend PostgreSQL, PostGIS, Git, GitHub	Analysis & Validation Stratified K-Fold, class imbalance, threshold tuning, error analysis

LEADERSHIP & ACTIVITIES

BITAmin Intercollegiate Big Data Club, 18th Cohort	Jul 2026 – Present
• Participate in deep-learning and multimodal-AI sessions, paper reviews, and team-project leadership	
EST AI Challengers, 1st Cohort	Jul 2026 – Present
• Lead AI/data projects and hackathons as team lead, project manager, and modeling lead	
TS Traffic Safety Supporters	Jul 2024 – Nov 2024
YASU Tutoring Program	Sep 2023 – Nov 2023

HONORS & QUALIFICATIONS

AWARDS	QUALIFICATIONS
Silver Award, Capstone Design Competition	Engineer Information Processing — Written Exam Passed (Aug 2026)
Portfolio Contest Award	ERP HR Level 2 · ITQ (3 modules) · Tableau Training

RESEARCH DIRECTION

I aim to develop industrial AI systems that move beyond detecting rare process anomalies to explaining model decisions and failure conditions in ways that field engineers can trust. My longer-term goal is to extend this work through process mining, multimodal integration of time-series, vision, and text data, and robustness evaluation under distribution shift.